

Lecture 08B: Difference-in-Differences and Event Studies

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Reminder: Natural Experiment Behind Our DD Example

- In Lecture 03, we studied the 2003 NO_x Budget Program (NBP)
- The NBP regulates summer pollution from power plants in some states but not others
- This creates a natural treatment-control comparison:
 - Treated units: NBP states in summer
 - Control units: non-NBP states and/or untreated seasons
- Our causal question was simple:

Did the policy actually reduce pollution?

Recall the Tennessee 2x2 DiD Logic

- To build intuition, we looked at a 2x2 table for Tennessee
- The four cells are:
 - Before / Winter
 - Before / Summer
 - After / Winter
 - After / Summer
- Summer is the “treatment” season because the NBP applies in summer
- Winter is the “control” (comparison) season because the NBP does *not* apply in winter
- The question: did summer pollution fall after 2003 by more than winter pollution?

Raw Difference-in-Differences: What Were We Computing?

$$\delta_{DD} = (\bar{Y}_{After}^{Summer} - \bar{Y}_{Before}^{Summer}) - (\bar{Y}_{After}^{Winter} - \bar{Y}_{Before}^{Winter})$$

- First difference: how much did summer emissions change?
- Second difference: how much did winter emissions change?
- Final difference: subtract off changes that would have happened *anyway*

Interpretation

DD asks whether the treated group changed more than the control group, from before the policy to after the policy.

Tennessee Example: The 2x2 Table

	Summer	Winter	Difference
Post-treatment (2003-2007)	27	59	-32
Pre-treatment (1997-2002)	77	71	6
Difference	-50	-12	-38

Source: Deschenes-Greenstone-Shapiro (AER 2017)

- This is the raw DD calculation in table form
- The entire regression framework we use next is just a more general way of writing this same object

How to Replicate the DD Table as a Regression

$$Y_{it} = \alpha + \beta \text{Treated}_i + \gamma \text{Post}_t + \delta (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

$$\bar{Y}_{C,\text{Pre}} = \alpha$$

$$\bar{Y}_{T,\text{Pre}} = \alpha + \beta$$

$$\bar{Y}_{C,\text{Post}} = \alpha + \gamma$$

$$\bar{Y}_{T,\text{Post}} = \alpha + \beta + \gamma + \delta$$

Each coefficient written with means

$$\alpha = \bar{Y}_{C,\text{Pre}}, \quad \beta = \bar{Y}_{T,\text{Pre}} - \bar{Y}_{C,\text{Pre}}, \quad \gamma = \bar{Y}_{C,\text{Post}} - \bar{Y}_{C,\text{Pre}}$$

$$\delta = (\bar{Y}_{T,\text{Post}} - \bar{Y}_{T,\text{Pre}}) - (\bar{Y}_{C,\text{Post}} - \bar{Y}_{C,\text{Pre}})$$

Another Quasi-Experiment: Denmark Tax Scheme (KLS AER 2013)

- Policy reform: after 1991, foreign workers in Denmark with incomes in the top bracket faced a preferentially low tax rate
- Treatment definition: treated workers are Denmark residents in the top bracket eligible for the preferential rate; control workers are Denmark residents just below that bracket
- Design: compare total foreign workers across these tax brackets before versus after 1991
- 1995 Bosman reform: European Court of Justice removed barriers to soccer player migration (e.g., nationality quotas on clubs)
- This setup is designed to recover migration responses to tax rates

Constructing the Standard DD Table

	Pre	Post
Treated ($Treated = 1$)	$\bar{Y}_{T,Pre}$	$\bar{Y}_{T,Post}$
Control ($Treated = 0$)	$\bar{Y}_{C,Pre}$	$\bar{Y}_{C,Post}$

- Start by comparing foreign-worker outcomes across tax brackets in pre and post periods

$$\delta_{DD} = (\bar{Y}_{T,Post} - \bar{Y}_{T,Pre}) - (\bar{Y}_{C,Post} - \bar{Y}_{C,Pre})$$

- First difference: treated pre-to-post change
- Second difference: control pre-to-post change
- Final difference: the treated change net of the control change

Replicating the DD Table Estimate Using a Regression

$$\delta_{DD} = (\bar{Y}_{T,Post} - \bar{Y}_{T,Pre}) - (\bar{Y}_{C,Post} - \bar{Y}_{C,Pre})$$

- Running the following regression exactly replicates δ_{DD}
- Key: You can add control variables to the regression, and δ will give you the DD effect controlling for other variables

$$Y_{it} = \alpha + \beta Treated_i + \gamma Post_t + \delta (Treated_i \times Post_t) + \varepsilon_{it}$$

$$\delta = \delta_{DD}$$

Go to the Data

The DD Table and the Replicating Regression

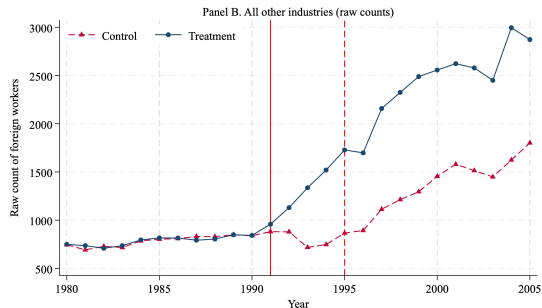
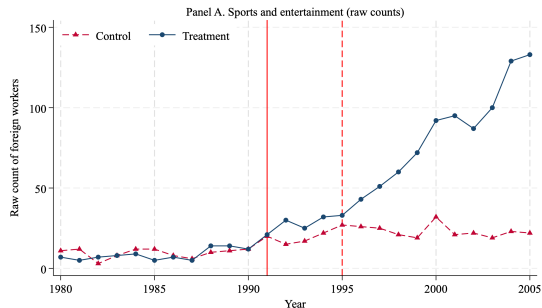
	Pre	Post
Treated (top bracket)	8.45	66.87
Control (below bracket)	9.55	22.07
DiD coefficient	45.89	

```
. reg count_workers D_treat D_post D_treat_post, vce(robust)
```

```
Linear regression               Number of obs   =       52
                                F(3, 48)         =      44.66
                                Prob > F          =     0.0000
                                R-squared         =     0.6070
                                Root MSE      =     20.457
```

count_work~s	Robust					
	Coefficient	std. err.	t	P> t	[95% conf. interval]	
D_treat	-1.090909	1.35627	-0.80	0.425	-3.817873	1.636055
D_post	12.52121	1.411722	8.87	0.000	9.682755	15.35967
D_treat_post	45.89091	9.877795	4.65	0.000	26.03027	65.75155
_cons	9.545455	.8912857	10.71	0.000	7.753405	11.3375

Appendix: Raw Counts Figures



Appendix: Normalized Figures

